CertaRig Phase 3

Hardware Inventory, Evidence and Commissioning Register

Baseline: 06 September 2026 | Phase 3 deadline: 20 September 2026

17 / 17	0 / 17	1 open gate
Robu product lines visually mapped to uploaded photographs.	Items electrically accepted so far. Receipt is not a functional test.	A separate regulated 5 V, 1-2 A relay and indicator supply is not evidenced.

Control statement

This register reconciles the Robu invoice against the supplied photographs. It deliberately separates visual receipt from electrical acceptance and integrated test evidence. The source invoice contains personal contact and address information; those details have been intentionally excluded from this project-facing register.

Source	Supplier	Invoice date	Invoice reference	Hardware value	Invoice total
Robu_ItemsList_CertaRig.pdf	Robu.in / Macfos Limited	04 Sep 2026	INV2627/225826	INR 5,824.25 incl. tax	INR 6,072.25 incl. freight and COD

Safety boundary

Phase 3 Wave 1 is a low-voltage educational proof of concept. Do not connect the relay contacts to mains, pumps or a claimed 30 A load. The emergency-stop switch is a demonstration interlock until its two NC contacts are mapped and tested; it is not treated as a certified safety component.

1. Readiness decision and wave boundary

Proceed now	Hold outputs	Defer water loop
Pi provisioning, visual inspection, continuity tests and potentiometer characterisation can start immediately. The Amazon delivery is not a blocker for these activities.	Do not energise the relay or indicators until a separate regulated 5 V supply is available and the relay pinout, input polarity and fuse path are documented.	Pumps, reservoirs, hoses and real fluid sensors belong to Wave 2. Purchase only after the professor confirms they are required for the 20 Sep evidence.

Wave 1 - mandatory dry hardware-in-loop bench

Raspberry Pi 3 Model A+ running the edge service; two 10 kohm panel controls representing pressure and flow; ADS1115 at 3.3 V; dual-NC latching emergency stop; protected relay inputs; 1 A fused, separate 5 V low-voltage output branch; green and yellow indicators; dashboard and logs on the laptop over Wi-Fi.

Wave 2 - optional real open-reservoir water loop

A small low-voltage pump, two open reservoirs, tubing, clamps, leak tray, compatible flow and pressure sensors, return path and physical mounting. It demonstrates real fluid behaviour, but adds leakage, priming, electrical isolation and calibration risk. It should not displace the Wave 1 evidence unless explicitly required by the advisor.

Meeting question	Why it must be frozen on 7 Sep
Is the dry hardware-in-loop bench sufficient for Phase 3, or is a real water loop mandatory by 20 Sep?	Defines whether Wave 2 is purchased and built.
Which evidence is graded: report, repository, test logs, photos, live demo, and/or video?	Prevents spending the remaining time on the wrong artefact.
Are potentiometers accepted as pressure and flow sensor emulators for the Phase 3 prototype?	Confirms the stated abstraction.
Is Raspberry Pi 3 Model A+ an acceptable substitute for the unavailable Zero 2 W?	Records the compute-platform change.
Should the AI layer remain advisory while deterministic interlocks retain authority?	Freezes the safety and agentic-system boundary.
What minimum test cases and pass metrics should appear in the template?	Freezes acceptance criteria before testing.

2. Robu shipment reconciliation

Status vocabulary: Received - visual match means the item or exact package code is visible. It does not mean electrically accepted or safe to energise.

ID	Robu code	Item	Qty	Role	Photo	Wave	Current status
R01	1364532	Green 3-9 V, 6 mm metal indicator with 15 cm cable	1	Healthy / permitted-state indicator	IMG_1837	Wave 1	Received - visual match
R02	1364728	Yellow 3-9 V, 8 mm metal indicator with 15 cm cable	1	Warning / inhibited-state indicator	IMG_1837	Wave 1	Received - visual match
R03	24438	MB102 830-point solderless breadboard	1	Low-voltage prototype interconnect	IMG_1850	Wave 1	Received - visual match
R04	R185776	Rubycon 100 uF, 50 V radial electrolytic capacitor	1	Local low-voltage bulk decoupling	IMG_1841	Wave 1 support	Received - visual match
R05	1897057	BF-013A 5 x 20 mm fuse holder	1	Fused low-voltage actuator branch	IMG_1839	Wave 1	Received - visual match
R06	R257326	24 AWG silicone wire - red	1	Labelled positive low-voltage wiring	IMG_1842	Wave 1	Received - visual match
R07	1825005	24 AWG flexible silicone wire - green	2	Labelled signal / interconnect wiring	IMG_1842	Wave 1	Received - visual match
R08	R165252	LANBOO LB16SM 16 mm latching emergency-stop switch, stated 2C-2NC	1	Two independent low-voltage inhibit / sense loops	IMG_1846	Wave 1	Received - visual match
R09	R195523	ALPS RK09K1130A5R 10 kohm rotary potentiometer	3	Breadboard calibration controls and spares	IMG_1838	Wave 1 support	Received - visual match
R10	R122767	Littelfuse 0217001.MXP, fast-acting 1 A, 250 V, 5 x 20 mm	5	Low-voltage branch protection	IMG_1845	Wave 1	Received - visual match
R11	662929	100 nF, 50 V through-hole disc capacitor	8	Local logic / ADC decoupling	IMG_1840	Wave 1 support	Received - visual match
R12	R181433	Murata 22 pF, 50 V, 0402 C0G SMD capacitor	6	Parts stock; not required for the Phase 3 bench	IMG_1844	Deferred	Received - visual match
R13	43582	ADS1115 16-bit, 4-channel I2C ADC module	2	Two analogue sensor-emulation channels; one spare module	IMG_1836	Wave 1	Received - headers loose
R14	R260139	2-channel 5 V relay module with optocouplers and guide rail	1	Low-voltage output switching for indicators only	IMG_1849	Wave 1	Received - visual match
R15	1163662	TE 23ESA103MMF50AF 10 kohm panel potentiometer	3	Panel controls: pressure, flow and spare / fault injection	IMG_1835	Wave 1	Received - exact code visible
R16	169868	Raspberry Pi 3 Model A+	1	Edge computer, deterministic interlocks, logs and API	IMG_1847, IMG_1848	Wave 1	Received - board and model confirmed
R17	R224115	BC547-TA NPN transistor	3	Protected relay-input interface / spares	IMG_1843	Wave 1 support	Received - exact code visible

Reconciliation result: all 17 invoiced physical product lines are represented in the 16 uploaded photographs. The invoice quantity of 43 includes 41 physical product units plus one freight and one COD-charge line.

3. Electrical acceptance register

Complete each row only after a measured test. Record the date, instrument or command, measured result, pass/fail and evidence filename in the generated Markdown companion or the next revision of this PDF.

ID	Component	Acceptance check	Status at baseline	Evidence to retain
R01	Green 3-9 V, 6 mm metal indicator with 15 cm cable	Polarity and current test	NOT TESTED	Photo + reading / terminal output
R02	Yellow 3-9 V, 8 mm metal indicator with 15 cm cable	Polarity and current test	NOT TESTED	Photo + reading / terminal output
R03	MB102 830-point solderless breadboard	Rail continuity and split-rail map	NOT TESTED	Photo + reading / terminal output
R04	Rubycon 100 uF, 50 V radial electrolytic capacitor	Capacitance / condition; observe polarity	NOT TESTED	Photo + reading / terminal output
R05	BF-013A 5 x 20 mm fuse holder	Continuity and mechanical retention	NOT TESTED	Photo + reading / terminal output
R06	24 AWG silicone wire - red	Inspect insulation; label both ends	NOT TESTED	Photo + reading / terminal output
R07	24 AWG flexible silicone wire - green	Inspect insulation; do not rely on colour alone	NOT TESTED	Photo + reading / terminal output
R08	LANBOO LB16SM 16 mm latching emergency-stop switch, stated 2C-2NC	Map both NC contact pairs with multimeter	NOT TESTED	Photo + reading / terminal output
R09	ALPS RK09K1130A5R 10 kohm rotary potentiometer	End-to-end resistance and smooth wiper sweep	NOT TESTED	Photo + reading / terminal output
R10	Littelfuse 0217001.MXP, fast-acting 1 A, 250 V, 5 x 20 mm	Continuity; confirm fit in holder	NOT TESTED	Photo + reading / terminal output
R11	100 nF, 50 V through-hole disc capacitor	Visual condition; install close to load	NOT TESTED	Photo + reading / terminal output
R12	Murata 22 pF, 50 V, 0402 C0G SMD capacitor	No bench test required	NOT TESTED	Photo + reading / terminal output
R13	ADS1115 16-bit, 4-channel I2C ADC module	Solder header, 3.3 V power, I2C scan, stable readings	NOT TESTED	Photo + reading / terminal output
R14	2-channel 5 V relay module with optocouplers and guide rail	Document pins; test coils from separate 5 V supply	NOT TESTED	Photo + reading / terminal output
R15	TE 23ESA103MMF50AF 10 kohm panel potentiometer	End-to-end resistance, wiper sweep, terminal map	NOT TESTED	Photo + reading / terminal output
R16	Raspberry Pi 3 Model A+	Boot, power stability, Wi-Fi, SSH, GPIO and I2C	NOT TESTED	Photo + reading / terminal output
R17	BC547-TA NPN transistor	Identify E-B-C pinout; diode-test junctions	NOT TESTED	Photo + reading / terminal output

4. Existing support inventory and remaining procurement gate

Support item	Baseline status
32 GB Class 10 A1/A2 microSD card	Available
USB microSD card reader	Available
Official 5 V, 2.5 A micro-USB Raspberry Pi supply	Available
Male-male, male-female and female-female Dupont jumpers	Available
MB102 power module or suitable supply breakout	Available
Additional black and blue/yellow 24 AWG wire	Available
2N2222 transistors	Available; BC547 units also received
2.54 mm male header strip	Available; ADS1115 headers also pictured loose
Screw terminals or lever connectors	Available
Heat-shrink tubing, cable ties and labels	Available
Rigid non-conductive mounting plate or ABS enclosure	Available
Digital multimeter	Available
Wire tools, screwdrivers and soldering equipment	Available
Separate regulated 5 V, 1-2 A supply for relay and indicators	NOT EVIDENCED - confirm or obtain before output test

Amazon delivery status

Item	Status	Phase 3 decision
ELEGOO 17-value 1% resistor assortment	Pending Amazon	Useful for 1 kohm base resistors, 10 kohm pull-downs and indicator current limiting. Does not block Pi-only setup.
10 uF, 16 V electrolytic capacitors, pack of 8	Pending Amazon	Optional local bulk decoupling; the received 100 uF capacitor is sufficient for initial low-voltage bench checks.
1000 uF, 25 V electrolytic capacitors, pack of 10	Pending Amazon	Not required for Wave 1. Hold for later supply-transient experiments or Wave 2.

Blocking decision

Confirm or obtain one regulated 5 V, 1-2 A supply dedicated to the relay coils and indicators. A reputable 5 V USB adapter or power bank can be used only with a secure breakout and measured output. Keep the Raspberry Pi on its official supply. Do not infer isolation from the word optocoupler; first document the relay module VCC, JD-VCC, GND and input arrangement.

5. Photographic evidence catalogue

Each photo is mapped by filename to the inventory ID. Packaging visibility is evidence of receipt, not proof of rating, authenticity or electrical performance.

IMG_1835 | R15
Three TE 23ESA panel potentiometers
Exact Robu code 1163662 is visible on all three bags.

IMG_1836 | R13
Two ADS1115 ADC modules
Exact Robu code 43582 is visible. Loose header strips are present and must be soldered.

IMG_1837 | R01, R02
Green and yellow metal indicators
Exact Robu codes 1364532 and 1364728 are visible.

IMG_1838 | R09
Three ALPS RK09 10 kohm potentiometers
Exact Robu code R195523 and quantity 3 are visible.

5. Photographic evidence catalogue - continued

IMG_1839 | R05
BF-013A fuse holder
Exact Robu code 1897057 is visible.

IMG_1840 | R11
Eight 100 nF disc capacitors
Exact Robu code 662929 and quantity 8 are visible.

IMG_1841 | R04
Rubycon 100 uF radial electrolytic capacitor
Exact Robu code R185776 and quantity 1 are visible.

IMG_1842 | R06, R07
Red and green 24 AWG silicone wire
Exact Robu codes R257326 and 1825005 are visible.

5. Photographic evidence catalogue - continued

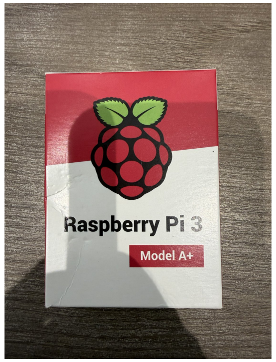
IMG_1843 | R17
Three BC547-TA NPN transistors
Exact Robu code R224115 and quantity 3 are visible.

IMG_1844 | R12
Six complimentary 22 pF 0402 SMD capacitors
Exact Robu code R181433 and quantity 6 are visible.

IMG_1845 | R10
Five Littelfuse 1 A fast-acting fuses
Exact Robu code R122767 and quantity 5 are visible.

IMG_1846 | R08
LANBOO latching emergency-stop switch
Exact Robu code R165252 is visible; electrical contact arrangement remains to be measured.

5. Photographic evidence catalogue - continued



IMG_1847 | R16
Raspberry Pi 3 Model A+ retail box
Model identity is clearly visible.

IMG_1848 | R16
Raspberry Pi 3 Model A+ board
Board marking and Robu code 169868 are visible.

IMG_1849 | R14
Two-channel 5 V optocoupled relay module
Exact Robu code R260139 is visible.

IMG_1850 | R03
MB102 solderless breadboard
Product format and MB-102 marking are visible.

6. Commissioning workflow - evidence before integration

Gate	Action	Pass condition	Do not proceed if
A - Quarantine	Keep all outputs unpowered; inspect for damage and label every bag / board.	No visible damage; exact ID retained.	Cracked board, bent connector, exposed conductor or uncertain identity.
B - Passive checks	Test fuses, holder, both E-stop NC circuits, pot end-to-end resistance / wiper sweep and transistor junctions.	Measurements match expected behaviour and are recorded.	Contact state is ambiguous, pot is open/intermittent, or fuse does not seat.
C - Pi only	Flash Raspberry Pi OS Lite 64-bit, boot on official supply, configure Wi-Fi / SSH, record OS and IP.	Stable boot and 20-minute idle run; no undervoltage report.	Power warning, thermal issue, corrupt card or unstable network.
D - ADC input	Solder ADS1115 header; power from Pi 3.3 V; scan I2C; connect one pot, then two.	Address detected; raw values move monotonically; no out-of-range voltage.	Header bridge, wrong rail, missing ground or unstable samples.
E - Software safety	Pass unit tests for E-stop polarity, broken wire, stale data, raw range, restart default and relay feedback mismatch.	All safety tests pass with outputs de-energised on every fault.	Any fault can energise or retain an output.
F - Output branch	With Pi isolated from coil power as designed, use separate 5 V supply, transistor interface, flyback protection if not on module, 1 A fuse and indicators only.	Each channel follows command and de-energises on E-stop / process fault.	Relay input current / polarity is unknown or the separate supply is absent.
G - Integrated proof	Run named test cases and 60-minute soak; collect CSV, screenshots, photos and commit ID.	All pass criteria and recovery rules are evidenced.	Results exist only as narration or simulation.

Critical software gate

Before any relay is energised, make the E-stop and feedback polarities configurable and verify the intended dual-NC fail-safe behaviour on the actual GPIO wiring. Add raw ADC guard bands, stale-sample detection, feedback mismatch detection and a safe de-energised start state. The AI layer may recommend or explain; it must not bypass deterministic interlocks.

7. Minimum evidence package for 20 September

Evidence group	Minimum artefact	Acceptance statement
Physical build	Dated overview and close-up photographs; labelled wiring diagram; component map.	A reviewer can identify Pi, ADS1115, two controls, E-stop, relay, fuse and indicators.
Pi deployment	Fresh-install commands, service status, OS / Python versions and network/API proof.	The PoC runs on Raspberry Pi hardware, not only the laptop simulator.
Analogue path	I2C scan; raw and converted values; three-point calibration for two channels.	Both channels are monotonic, bounded and repeatable within the declared tolerance.
Safety path	E-stop normal/pressed, one broken NC wire, startup/restart, stale sample and ADC raw-range tests.	Every unsafe or uncertain state de-energises outputs and records a reason.
Output path	Two relay-channel tests using low-voltage indicators plus commanded-versus-feedback evidence.	Outputs cannot remain energised when the safety decision is false.
Reliability	A 60-minute timestamped soak log and summary of errors / restarts.	No unexplained output transition; failures are disclosed rather than hidden.
Repository	Public commit/tag, README, wiring, configuration example, tests, deployment script and evidence folder.	A fresh clone can reproduce the software and locate the hardware evidence.
Report / demo	Phase 3 template completed only with achieved results, limitations, repository link and any requested video link.	Claims match recorded evidence and the frozen advisor scope.

Suggested named test cases

TC01 normal operating range; TC02 warning threshold; TC03 trip threshold; TC04 E-stop press and latch; TC05 one NC wire disconnected; TC06 stale ADC sample; TC07 raw ADC below/above guard band; TC08 relay feedback mismatch; TC09 process restart with fault present; TC10 network loss while edge interlock remains active.

8. Schedule and change-control baseline

Date	Milestone	Deliverable
6 Sep	Receive and control	Freeze inventory, quarantine untested parts, inspect Pi and relay, identify pin labels, prepare microSD.
7 Sep	Professor scope freeze	Confirm whether a hardware-in-loop dry bench is sufficient for Phase 3, and whether the water loop is explicitly required.
8-9 Sep	Component acceptance	Continuity tests, potentiometer sweeps, transistor diode tests, solder ADS1115 headers, Pi boot and I2C detection.
10 Sep	Software safety gate	Correct E-stop and feedback polarity configuration, add stale-sample and raw-range faults, and pass automated tests before outputs.
11-12 Sep	Input integration	Wire two panel potentiometers to ADS1115 at 3.3 V, collect raw data, perform 3-point calibration and log evidence.
13-14 Sep	Output and safety integration	Use a separate regulated 5 V supply, transistor interface and 1 A fused branch. Drive only low-voltage indicators.
15 Sep	Integrated tests	Run normal, threshold, E-stop, broken-wire, stale-data, relay-feedback and restart tests; then perform a 60-minute soak.
16 Sep	Evidence capture	Photograph wiring, export CSV logs, screenshots and test results; record exact software commit and configuration.
17-18 Sep	Phase 3 report	Populate the template only with achieved evidence, limitations and measured results; advisor review on 18 Sep.
19 Sep	Submission rehearsal	Fresh clone, clean installation, signed-out link check, PDF review and buffer for corrections.
20 Sep	Submit	Submit the verified report, repository tag, evidence package and video/link set requested by the professor.

How to update this register

Use the inventory ID as the permanent key. When an item is tested, append the date, measured value, pass/fail result and evidence filename. When the hardware changes, create a new dated revision rather than overwriting the baseline. Never mark an item accepted solely because the package label matches.

Revision	Date	Change	Evidence / approver
1.0	06 Sep 2026	Initial Robu shipment reconciliation, 16-photo mapping and Phase 3 commissioning baseline.	Robu invoice + IMG_1835 to IMG_1850
1.1			
1.2			

Immediate decision

Start passive acceptance and Raspberry Pi provisioning now. The only procurement gate for Wave 1 output testing is the separate regulated 5 V supply. Do not purchase Wave 2 water-loop hardware until the 7 September meeting freezes the deliverable.